Journal of Electromyography and Kinesiology 25 (2015) 522–530

Contents lists available at ScienceDirect

Journal of Electromyography and Kinesiology

journal homepage: www.elsevier.com/locate/jelekin

Lower extremity muscle activation and function in progressive task-oriented training on the supplementary tilt table during stepping-like movements in patients with acute stroke hemiparesis

Chang-Yong Kim [a,1], Jung-Sun Lee [b,2], Hyeong-Dong Kim [c,*], Junesun Kim [c,3], In-Hak Lee [d,4]

[a] Department of Health Science, The Graduate School, Korea University, Jeongneung 3-Dong, Seongbuk-Gu, Seoul 136-703, Republic of Korea
[b] Department of Epidemiology and Health Informatics, The Graduate School of Public Health, Korea University, Anam-Dong, Seongbuk-Gu, Seoul 136-703, Republic of Korea
[c] Department of Physical Therapy, College of Health Science, Korea University, Jeongneung 3-Dong, Seongbuk-Gu, Seoul 136-703, Republic of Korea
[d] Department of Physical Therapy, Daejeon Health and Sciences College, 77-3, Gayang 2-Dong, Dong-Gu, Daejeon 300-711, Republic of Korea

ARTICLE INFO

Article history:
Received 9 October 2014
Received in revised form 5 February 2015
Accepted 21 March 2015

Keywords:
Acute stroke
Electromyography
Functional performance
Task-oriented training
Tilt table

ABSTRACT

An effective and standardized method for applying a tilt table as a supplementary treatment in the early rehabilitation of stroke patients is still missing. The aim of this study was to determine the influence of progressive task-oriented training on the tilt table on the improvement in lower extremity (LE) muscle activation and clinical function in subjects with hemiplegia due to stroke. Thirty-nine subjects with acute stroke were randomly allocated to three groups; control group, tilt table group, and task-oriented training group on the tilt table, with 13 patients, respectively. All of the subjects received the routine therapy for half an hour, and subjects in the experimental groups additionally received training on two different tilt table applications for 20 min a day, five times a week for three weeks. The effect of tilt table applications was assessed using the surface electromyography (EMG) analysis during stepping-like movements on the tilt table for LE muscle activation and clinical scores for function. Our results showed that there was a significantly greater increase in the EMG patterns of the extensors and flexors of the affected leg muscles during flexion and extension movements of both legs and clinical scores in patients undergoing the progressive task-oriented training on the tilt table compared to the other groups. These findings suggest that progressive task-oriented training on the tilt table can improve LE muscle activation and clinical scores of functional performance for early rehabilitation of subjects with acute stroke.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

Acute stroke patients are bedridden immediately after brain damage, and therefore, lack of adequate rehabilitation may lead to irreversible anatomical and functional changes, as well as progressive weakness (Solopova et al., 2011a,b). Since these musculoskeletal instabilities and the problem of motor neurorehabilitation after stroke also lead to delayed functional training in these patients (Czell et al., 2004), it is important that the patients are progressively mobilized and should undergo supported-weight load training during the early stages of rehabilitation (Browse, 1965).

Stroke rehabilitation is an organized effort to help patients maximize all opportunities for returning to an active lifestyle (Aichner et al., 2002). During the past few decades, major progress in neuroscience has resulted in novel concepts for rehabilitation interventions after stroke. Various studies and systematic reviews support the efficacy of task-oriented training (Nudo, 2007; Rensink et al., 2009). Previous studies of task-oriented training already showed benefits for functional outcomes compared with traditional therapies. Task-oriented training includes a wide range of interventions such as walking training on the ground, treadmill training, target-matched reaching or kick training (Shumway Cook and Woollacott, 2007), and one-leg standing training for improving balance (Rensink et al., 2009). Such training is task and patient focused and not therapist focused.

Furthermore, in the clinical setting, most therapists use manual therapy such as neuro-developmental techniques or supplementary

* Corresponding author. Tel.: +82 2 940 2835; fax: +82 2 940 2830.
E-mail addresses: dreampt@korea.ac.kr (C.-Y. Kim), jssmile89@daum.net (J.-S. Lee), hdkimx1234@daum.net (H.-D. Kim), junokim@korea.ac.kr (J. Kim), ihlee@hit.ac.kr (I.-H. Lee).
[1] Tel.: +82 10 9807 0113; fax: +82 2 940 2830.
[2] Tel.: +82 10 3115 6119; fax: +82 2 2286 1175.
[3] Tel.: +82 2 940 2835; fax: +82 2 940 2834.
[4] Tel.: +82 42 670 9334; fax: +82 42 670 9330.

http://dx.doi.org/10.1016/j.jelekin.2015.03.004
1050-6411/© 2015 Elsevier Ltd. All rights reserved.

tilt table and cycle ergometer to increase the mobility of lower extremities (LE) and to perform supported-weight load training in patients (Czell et al., 2004). Among these, the tilt table is a method that has been used for over 60 years by physiologists and physicians for many purposes. These include the study of the human body's heart rate and blood pressure adaptations due to changes in position, for modeling responses to hemorrhage, as a technique for evaluating orthostatic hypotension, as a method for studying hemodynamic and neuroendocrine responses in congestive heart failure, autonomic dysfunction and hypertension, as well as a tool for drug research (Furlan et al., 2000; Hainsworth and Al-Shamma, 1988; Kautz and Brown, 1998). It has also become a useful device in the mobilization of traumatic brain injured and spinal cord injured patients, as well as in patients suffering from acute stroke, under the supervision of physical therapists (Wagner, 1959). In particular, the position of acute stroke patients can be changed continuously from horizontal to vertical using the tilt table during the early stage of rehabilitation. As a result, the patients can adapt to the state of walking for training just prior to it, and muscle atrophy and weakness can also be prevented (Czell et al., 2004).

However, a more effective and standardized method for applying the tilt table as a supplementary treatment in the early rehabilitation of stroke patients has not been presented. The tilt table consists of thoracic, pelvic, and knee safety belts which are connected to the sides of the table and are used to prevent forward tipping as well as to fix the patient's body while they are standing or leaning against it. The tilt table was adapted by physical therapists and its angle was gradually increased (Luk'ianov et al., 2010; Solopova et al., 2011a,b). In general, during the application of the tilt table in the clinical setting, the patients are strapped by knee belts on both the affected and less-affected LE, which leads to delay in the proprioceptive input and muscular activity of the affected LE (Riberholt et al., 2013). Actually, more than half of the patients are unable to walk in the acute period and develop an abnormal movement stereotype during stepping-like walking (Solopova et al., 2011a,b). There are no studies investigating muscle activation changes in the LE after employing the tilt table during stepping-like movements in acute stroke patients. Based on the above background, this study aimed to identify whether tilt tables are more effective in increasing muscle activity and to investigate whether progressive task-oriented training on the tilt table by applying a knee belt in a different manner can improve LE muscle activation for early rehabilitation of acute stroke patients. We also evaluated the impact of the degree of performance on activities of daily living and LE function using clinical measurements. Therefore, this study established a hypothesis that tilt table and progressive task-oriented training on the tilt table would increase muscle activity from the LE, which will improve LE functions following acute stroke. We also predicted that progressive task-oriented training on the tilt table would be more effective than the tilt table alone.

2. Methods

2.1. Subjects

Thirty-nine subjects (21 women, 18 men) with first-ever acute stroke from the stroke rehabilitation institute participated in this study and gave their informed consent. Sample size estimate was based on data collected from previous studies (Kuznetsov et al., 2013; Solopova et al., 2011a,b). A priori power analysis determined that a sample size of 13 subjects with acute stroke in each group was required to obtain a statistical power of 0.80 using the General power analysis program (GPower 3.1) (Faul et al., 2007). This was based on one-way ANOVA measurements of the comparison among the three groups, with a predetermined reliability coefficient of 0.9 (Eliasziw et al., 1994). Stroke diagnosis and location of lesions were based on computed tomography or magnetic resonance imaging as well as neurological functions. Subjects were selected for inclusion if they met the following conditions: (1) they had stable hemodynamics in the absence of significant lower limb spasticity (with Ashworth index <2 in all of the lower limb muscles: average 0.9 ± 0.6 [mean ± SD]) and absence of significant cardiovascular impairment, (2) an ischemic or hemorrhagic post-stroke hemiparesis, (3) ⩾26 on the modified mini-mental status examination Korea version (MMSE-K), and (4) stable medical condition allowing participation in the experimental procedures and intervention. Subjects were excluded from the study if they had cardiac arrhythmia, thrombophlebitis, significant perceptual, cognitive, or communication impairments, diabetes, unilateral neglect, and contraindication to tilt table and surface electromyography (sEMG) (cancer, pacemaker, unstable epilepsy, or skin abnormalities). Unilateral neglect was tested using the Star Cancellation Test of visuospatial neglect. Those scoring less than 47 were excluded from participation.

2.2. Outcome measures

Changes in muscle activation and functional performance of the LE were evaluated using sEMG analysis and clinical measurements, which were performed before and after the three-week intervention period. Three physical therapists blinded to the group allocation performed the evaluations. Prior to performing clinical measurements, these blinded evaluators underwent an 8-h training session. Rater competency was assessed by the primary investigator with nine years of experience in performing these measurements. The evaluators were trained to conduct the sEMG analysis in accordance with the standardized procedures as described below. Subjects were advised not to disclose their treatment assignment to the evaluator.

2.2.1. Surface electromyography analysis

sEMG analysis showed muscle activation changes in the LE during stepping-like movements on the tilt table, which was inclined by 30°, while all subjects in each group were strapped to the table using only the safety thoracic and pelvic belts (Solopova et al., 2011a,b). To assess their abilities to execute "locomotor-like" movements on the tilt table, the subjects were asked to perform alternating active flexion–extension movements of both legs at a comfortable speed as quickly as they could during the pre and post measurements. Overall, three trials were recorded with a 1 min rest period between trials. The duration of each trial was approximately 60 s. We also measured limb load by recording the pressure distribution of the plantar surfaces of both feet using two foot switch sensors with insoles (Noraxon, USA, Inc., Scottsdale, AZ, USA). Two foot switch sensors with insoles were used to determine the foot-off time (converted to 0) and foot-contact time (converted to 1). The measurements were split into flexion and extension phases for both legs (Maegele et al., 2002). Signals from switch sensors were interpolated from 50 to 1000 Hz and synchronized with EMG signals. In addition, the subjects were allowed to wear shoes made of the same material (synthetic leather). Fig. 1 shows an example of the EMG pattern during stepping-like movements on the tilt table in one representative subject of each group. A Myosystem 1400A unit (Noraxon, USA, Inc., Scottsdale, AZ, USA) was used to measure the activation of biceps femoris (BF), medial gastrocnemius (MG), rectus femoris (RF), and tibialis anterior (TA) muscles in both legs according to SENIAM guidelines (Criswell, 2010; Hermens et al., 1999). Samples were collected at 1 kHz into a personal computer system (2.27 GHz processor, 2.00 GB RAM). The ground electrode was placed over the anterior superior iliac spine. Raw sEMG signals were band-pass filtered between 10 and

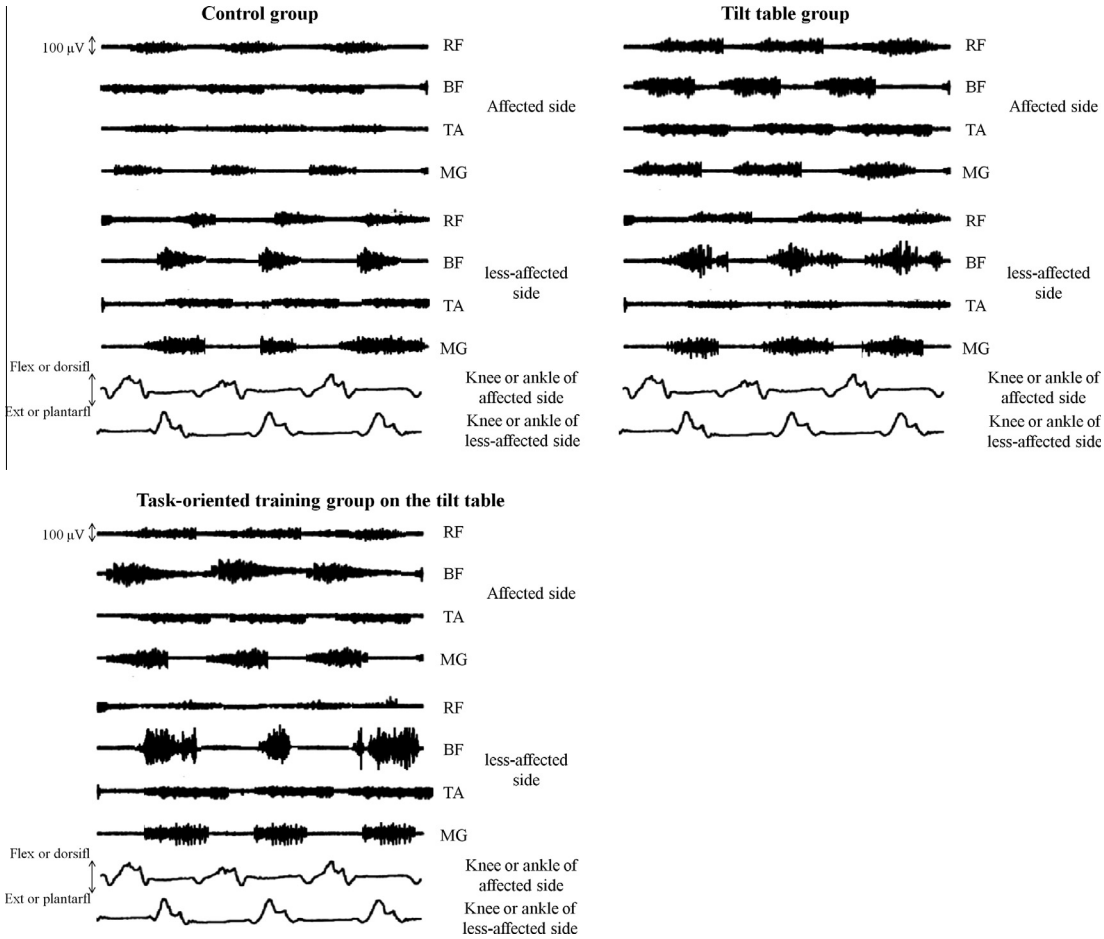


Fig. 1. Representative examples of EMG motor pattern in each group during stepping-like movements (3 consecutive steps) on the tilt table after the 3-week intervention. The last 2 signals below the raw EMG signals in each graph indicate the root-mean-square (RMS) of the EMG amplitude. RF – rectus femoris; BF – biceps femoris; TA – tibialis anterior; MG – medial gastrocnemius.

500 Hz. The MyoResearch software (MyoResearch XP; Noraxon, USA Inc.) was used for data processing and analysis. Signals were recorded using pairs of circular Ag/AgCl surface bipolar electrodes (Tyco Healthcare Korea, Seoul, Korea) with an inter-electrode distance of 20 mm, oriented parallel to the muscle fibers. The skin was shaved and cleaned with alcohol prior to the electrode placement. To ensure precise placement of the electrodes during repeated pre and post measurements, the areas of electrode placement were marked using the attached electrodes. Electrode placement was also verified through a series of functional tests (Tennant et al., 2014). For assessment of EMG amplitudes, raw EMG signals were numerically rectified and then the root-mean-square (RMS) of the EMG amplitude was calculated for each trial. The manual muscle testing (MMT) was used to determine the maximum voluntary contraction (MVC), and the RMS of the EMG during MVC was used to normalize the RMS of the EMG recorded in the trials. These procedures for MVC of each muscle were completed according to the recommendations by Daniels and Worthingham (2003) and also used as a measure of muscle activation. MVC of knee flexors and extensors was performed in the supine position when the knee joint was fixed at 90° (hip angle 45° and ankle angle 80°). The subject was instructed to relax the contralateral extended LE as much as possible during MVC. Measurements in subjects started with the less-affected LE. The subject was given one attempt at the MVC for training before recordings. Overall, 12 trials were recorded, with 6 trials on the less-affected LE (3 for flexion, and 3 for extension) and 6 trials on the affected LE, with a 1 min rest period between the trials. The mean muscle activation during a 5 s period before the command to perform MVC was used as a baseline. Strong verbal encouragement was given during each trial. The maximum muscle activation value obtained in each of the three attempts was used to calculate the MVC. Also, EMG envelopes were assumed to be zero if the maximum value was less than 3 μV, which was considered to be the noise level (Solopova et al., 2011a,b; Sylos-Labini et al., 2014). These data were exported as text data into a custom-written Microsoft Excel program to determine the metric for measuring muscle activation: the ratio of muscle activation during stepping-like movements on the tilt table to that at the normal grade of MMT (% MVC). The muscle activation for each trial of muscles was averaged over 3 trials.

2.2.2. Clinical evaluations

The Barthel Index (BI) was used to determine the functional status of each patient (Wade, 2000). Stroke severity was assessed using the National Institutes of Health Stroke Scale (NIHSS) (Brott et al., 1989). Motor function of LE was evaluated with the Fugl-Meyer Scale (FMS) (Fugl-Meyer et al., 1975). The hemiparesis severity was evaluated with a 6-grade scale (McPeak, 1996): paralysis (0) – no leg movements and no voluntary muscle contractions could be detected; severe paresis (1) – partial muscle contraction but without joint movements; pronounced paresis (2) – full range of joint movements under partial limb loading; moderate paresis (3) – full range of joint movements under full limb loading; light paresis (4) – movements under additional limb loading but weaker than on the non-paretic side; normal muscle force production (5). We also monitored the ability of patients to perform

everyday life activities before and after the intervention: to sit, stand, move from the wheelchair to the bed, and walk along a short 10 m walkway with or without assistance.

2.3. Experimental procedure

Test procedures are summarized in Fig. 2. Group allocation of the subjects was determined using a randomization procedure, in which each subject drew a card from a basket containing three cards with markings 1 (control group; CG), 2 (tilt table group; TG), or 3 (task-oriented training group on the tilt table; TTG), without looking at the card. Consequently, the subjects and their therapists were not aware of the group allocation. The baseline characteristics of all subjects in each group are presented in Table 1. The data indicated that the groups had similar demographic characteristics. Over a 3-week period, all of the subjects received functional training according to the daily routine schedule in the clinical setting, which consisted of strengthening and stretching exercises of the limbs, postural control training in sitting and standing positions, therapist-guided techniques for facilitating normal movement, and simple forward stepping. They received the above routine therapy for half an hour, five times a week. Subjects in the TG and TTG additionally received training on the supplementary tilt table (Midland Manufacturing Co., Inc., Columbia, SC, USA) for 20 min a day while they were positioned such that they felt comfortable at a tilt angle. The two groups received training on the following tilt table applications: TG – Subjects were strapped by safety thoracic, pelvic, and both knee belts; TTG – While subjects were strapped by safety thoracic, pelvic, and the knee belts only on the affected side, they progressively performed task-oriented training such as one-leg standing or target-matched reaching, and kick a ball exercises with the less-affected LE (Rensink et al., 2009; Shumway Cook and Woollacott, 2007) (Fig. 3). At first, subjects in the TTG stood and leaned against the tilt table with their trunk restrained to prevent compensatory trunk movement. Corrective feedback was given if compensatory movements were observed. Other tasks were also used to minimize compensatory movements. The subjects then started with an easy task, such as one-leg standing or closer target-matched reaching, and kicking light load training. As they completed the easy task perfectly, they were allowed to perform increasingly difficult tasks, such as far target-matched reaching and kicking heavy load training. The therapist also determined the task level of each subject on the basis of the principle of progressive load (Kisner and Colby, 2007). For the target-matched reaching and kicking a ball training, familiar objects such as plastic balls were used that varied in size, shape, and weight (56–453 g). They were only allowed to reach and kick in the sagittal plane of the anterior–posterior direction. Training intensity was determined depending on the response of each subject, and training involved only the less-affected limb. Instructions were to move at a preferred speed and to increase that speed as training progressed. The subjects in the TTG performed a total of 5 sets, with 10 repetitions in a set. After each set, a one-minute resting time was allowed. The angle of the tilt table measured between the surface of the table and horizontal was varied from 0° to 90°. During the 20-min phase of the intervention, all of the subjects were placed in the supine position on the tilt table, and were allowed to gradually increase their maximum tilt angle and to reduce their tilt angle during a session if they felt light headed. If they experienced dizziness or nausea during the experimental procedures, we immediately stopped the experiment, and the subjects were allowed to rest in the supine position. Furthermore, we used a therapeutic foam roller (length 60 cm, width 15 cm) to prevent knee hyper-extension in the subjects.

2.4. Data processing and statistical analysis

Statistical analysis of the data was performed by using the SPSS 12.0 software (SPSS Inc., Chicago, IL, USA). The values in each group were expressed as means and standard deviations. Since samples

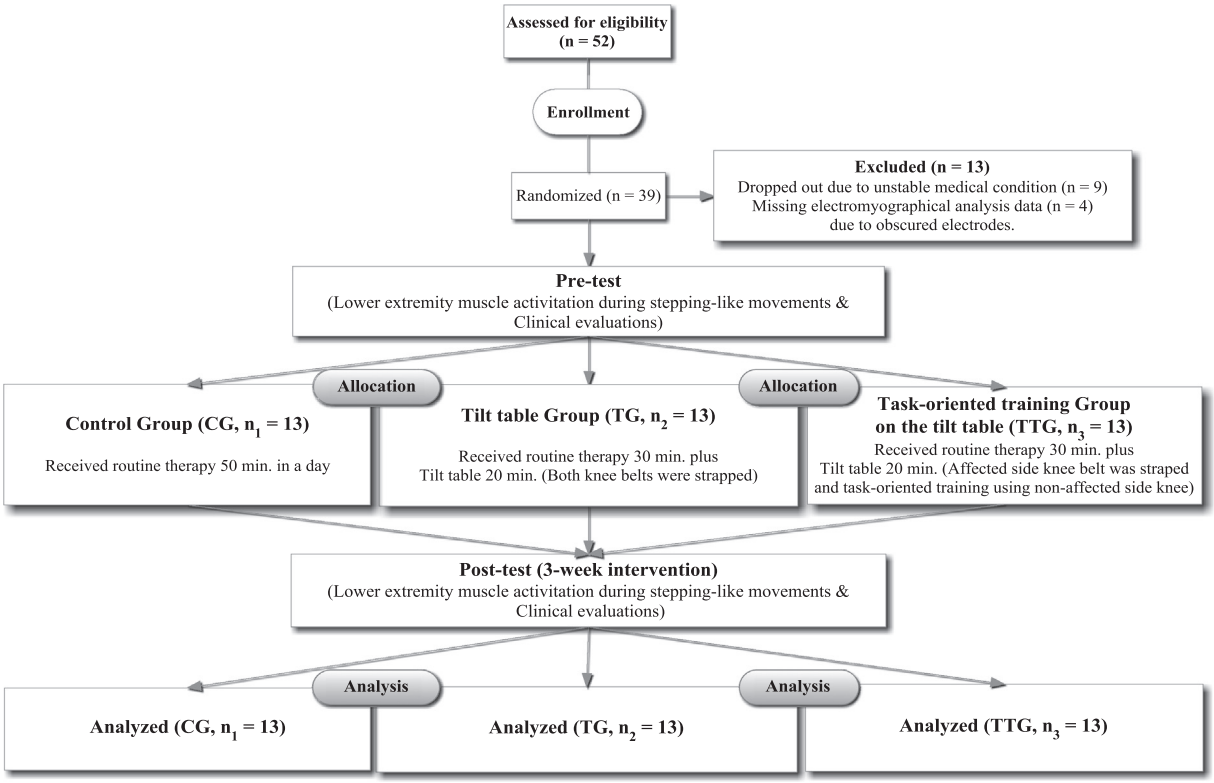


Fig. 2. Experimental procedures used in the study.

Table 1
Subject characteristics and clinical scores for each group.

Characteristics	Control group (CG, n_1 = 13)	Tilt table group (TG, n_2 = 13)	Task-oriented training group on the tilt table (TTG, n_3 = 13)	Task-oriented training group on the tilt table (TTG, n_3 = 13)	F/x^2 (p-Value[c])
Age (years)	58.54 ± 11.73	61.20 ± 8.72	59.53 ± 8.83	59.53 ± 8.83	−1.78 (.25)
Weight (kg)	66.53 ± 5.42	63.87 ± 5.14	62.59 ± 4.72	62.59 ± 4.72	−1.34 (.58)
Height (cm)	162.19 ± 5.02	163.57 ± 3.33	164.25 ± 4.28	164.25 ± 4.28	−2.74 (.07)
Days after stroke	23.71 ± 3.85	27.99 ± 4.32	26.12 ± 5.94	26.12 ± 5.94	−0.98 (.82)
Mini-mental state examination	26.91 ± 3.91	27.42 ± 4.28	28.42 ± 2.47	28.42 ± 2.47	0.57 (.38)
Star cancellation test (max = 54)	51.08 ± 0.97	50.08 ± 1.97	50.42 ± 2.71	50.42 ± 2.71	0.13 (.74)
Sex (male/female)	7/6[a]	5/8	6/7	6/7	0.95 (.87)
Ischemic/hemorragic stroke	4/9	6/7	8/5	8/5	−0.99 (.43)
Plegic side (right/left)	6/7	7/6	5/8	5/8	−1.21 (.76)
Clinical scores					
BI score					
Pre-test	16.91 ± 1.77	16.87 ± 2.11	17.13 ± 1.98	17.13 ± 1.98	−0.17 (.87)
Post-test	19.27 ± 5.97	70.28 ± 7.59[†]	76.99 ± 6.54[†,‡]	76.99 ± 6.54[†,‡]	103.14 (<.01)[**]
t (p-Value[b])	−3.18 (<.05)[*]	−14.53 (<.01)[**]	−15.42 (<.01)[**]	−15.42 (<.01)[**]	
NIHSS score					
Pre-test	9.21 ± 2.41	9.04 ± 1.88	9.84 ± 2.24	9.84 ± 2.24	−0.13 (.84)
Post-test	6.98 ± 2.01	4.11 ± 1.91[†]	4.21 ± 2.10[†]	4.21 ± 2.10[†]	10.22 (<.05)[*]
t (p-Value)	3.24 (<.05)[*]	9.89 (<.01)[**]	10.54 (<.01)[**]	10.54 (<.01)[**]	
FMS score					
Pre-test	9.28 ± 1.16	10.04 ± 1.08	10.95 ± 1.77	10.95 ± 1.77	−0.08 (.77)
Post-test	11.17 ± 1.78	21.99 ± 1.54[†]	28.27 ± 2.12[†,‡]	28.27 ± 2.12[†,‡]	37.55 (<.01)[**]
t (p-Value)	−4.23 (.08)	−11.42 (<.01)[**]	−13.21 (<.01)[**]	−13.21 (<.01)[**]	

Values are expressed as mean ± standard deviation.
[a] Values are expressed as numbers.
[b] Within-group comparison.
[c] Between-group comparison. BI – Barthel Index; NIHSS – National Institutes of Health Stroke Scale; FMS – Fugl-Meyer Scale.
[*] $p < .05$.
[**] $p < .01$.
[†] Significantly different compared to the CG.
[‡] Significantly different compared to the TG.

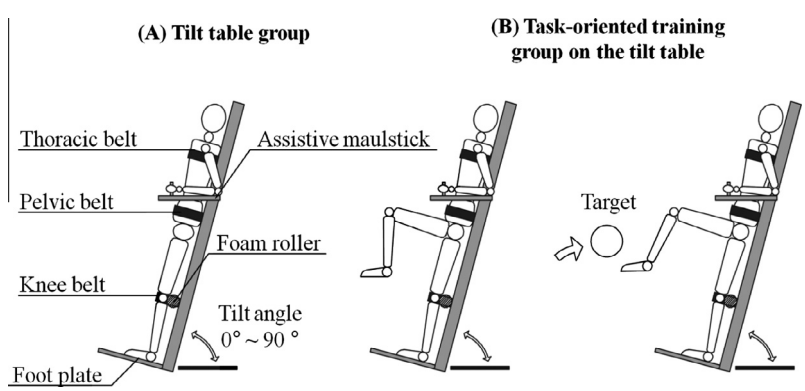


Fig. 3. Two different tilt table applications.

included in this study showed normal distribution curves in the Kolmogorov–Smirnov test, parametric methods were used. The chi-square test (gender, type of stroke, and side of hemiparesis) and one-way analysis of variance (ANOVA) (age, weight, height, duration, MMSE-K, and star cancellation test score) were used to compare demographic characteristics of subjects among the 3 groups. Comparisons of the clinical scores and muscle activation changes in the LE between the groups at the pre- and post-tests were investigated using repeated-measures one-way ANOVA, followed by the Bonferroni's post hoc test. Consequently, this test examined between-group differences (intervention time × group ANOVA). A post hoc test was performed to identify the locus of interactions. The paired t-test was performed to compare the same parameters before and after the intervention within each group. To assess the statistical significance of data collected from the subjects, the significance level was set at $p < 0.05$.

3. Results

3.1. Comparison of sEMG activation during stepping-like movements among groups

Fig. 4 illustrates changes in the mean EMG activation of LE muscles during flexion and extension phases of the affected and less-affected knee movements. During the post-test in the TG and TTG, there was significantly increased activation of the RF muscle ($p < 0.05$) in the TG and significantly increased activation of the

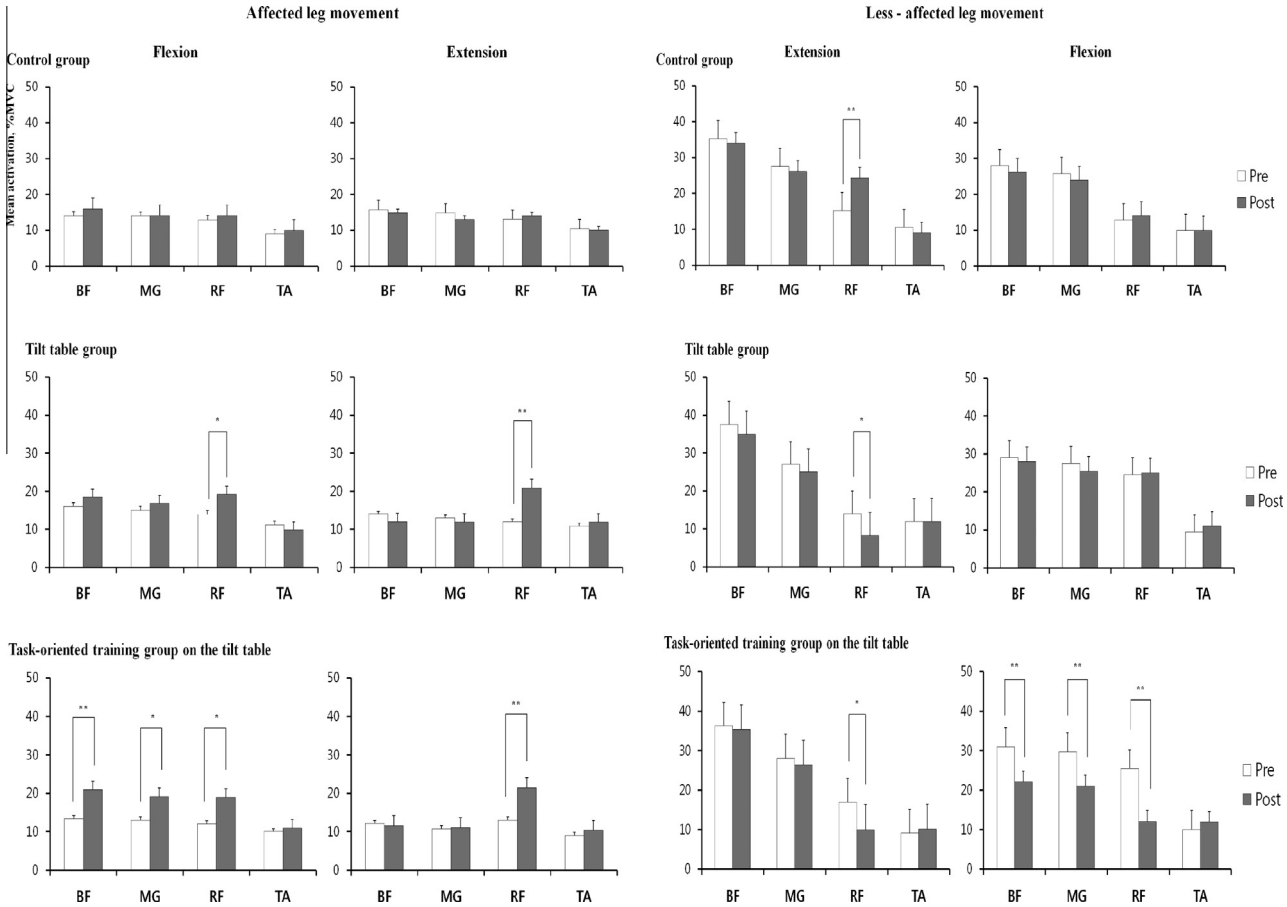


Fig. 4. EMG activation changes in the lower extremity (LE) during stepping-like movements before and after intervention in each group. The figure on the left side shows the mean activation (±SD) of the LE muscles during movements of the affected leg in each group. The figure on the right side shows the mean activation (±SD) of the LE muscles during movements of the less-affected leg in each group. BF – biceps femoris; MG – medial gastrocnemius; RF – rectus femoris; TA – tibialis anterior. Asterisks denote significant differences in mean muscle activation between pre and post intervention. $^{*}p < 0.05$, $^{**}p < 0.01$.

BF ($p < 0.01$), MG, and RF muscles of the affected side ($p < 0.05$) in the TTG. There was also significantly increased activation of the RF muscle of the less-affected side in the TG and TTG ($p < 0.05$) during flexion movement of the affected leg. On the other hand, there was significantly decreased activation of the BF, MG, and RF muscles ($p < 0.01$) of the less-affected side in the TTG, and significantly decreased activation of the RF muscle of the affected side in the TG and TTG ($p < 0.01$) during extension movement of the affected leg. During the post-test in the CG, there was significantly decreased activation of only the RF muscle ($p < 0.01$) of the less-affected side during flexion movement of the affected leg compared to the other groups.

Post-hoc testing revealed that in the TTG, the EMG activation changes in the BF and MG muscles of the affected side ($F = 14.892$ and 12.019, $p = 0.02$ and 0.03, respectively) during flexion movement of the affected leg were significantly increased compared to those in the CG and TG. The EMG activation changes in the BF, MG, and RF muscles of the less-affected side ($F = 18.775$, 21.484 and 27.783, $p = 0.01$, 0.01, and 0.002, respectively) during extension movement of the affected leg were significantly less than those in the CG and TG.

3.2. Comparison of clinical scores among groups

Table 1 shows the overall dynamics of changes in the clinical scores, and the improvement was significantly higher in the TTG. Especially, post hoc testing revealed that the BI and FMS scores

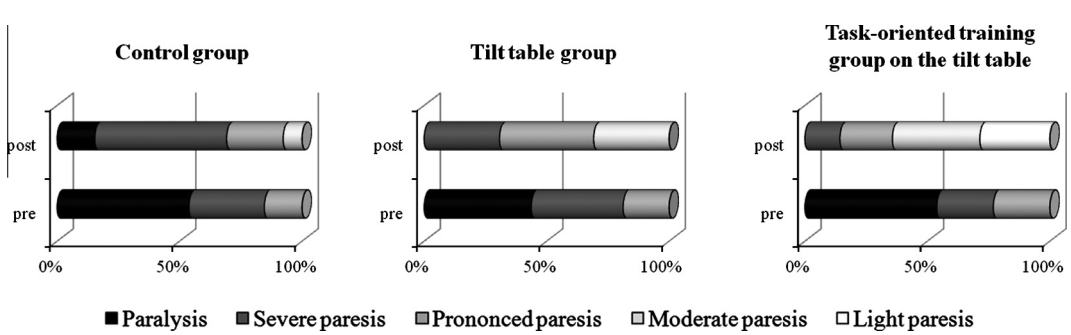


Fig. 5. The percentage of the degree of paresis for each group before and after intervention.

Table 2
Ability (number of subjects) to perform specific activities of daily living before and after intervention.

Activity	Control group (CG, n_1 = 13)		Tilt table group (TG, n_2 = 13)		Task-oriented training group on the tilt table (TTG, n_3 = 13)	
	Pre	Post	Pre	Post	Pre	Post
Assistance needed for movement from the wheelchair to the bed	11	5	12	3	10	1
Self-maintained sitting posture	2	4	1	6	3	9
Unaided standing	None	4	None	4	None	8
Unassisted walking in the ward	None	2	None	7	None	10

of the TTG subjects ($p < 0.01$) were significantly different from those of the CG and TG subjects.

In general, all of the subjects in the three groups showed improvements in their physical status, overall motor activity, and clinical scores. Especially, the TTG subjects had a more positive assessment than those in other groups. The reduction in the degree of hemiparesis is illustrated in Fig. 5. After intervention, 9 subjects (69%) in the TTG had light and moderate hemiparesis, while only 4 subjects (31%) in the TG and 1 subject (7%) in the CG had light and moderate hemiparesis.

Significant improvements in performance of everyday life activities were also observed in the TG and TTG subjects (Table 2). Prior to the intervention, subjects with pronounced paresis required assistance to stand with aids; however, following intervention, they could stand alone without any aids. Also, from 14 to 15 days after the initiation of intervention, they were able to walk over-ground with the help of one physical therapist. On the other hand, in the CG subjects, these abilities were relatively less developed and they either sat or just tried to stand near their bed with arm support. Therefore, a greater number of subjects in the TTG compared to those in the TG and CG were able to perform specific activities of daily living following intervention.

4. Discussion

This was the first study to compare specific protocols for the use of the tilt table in stroke rehabilitation in addition to comparing task oriented protocols against weight bearing alone. The study was conducted to investigate the clinical benefits of using supplementary progressive task-oriented training on the tilt table on the muscle activation changes in the LE, functional abilities, and degree of performing activities of daily living in patients with hemiparesis post-stroke. Rehabilitation with the help of this effective and standardized application of the tilt table strongly influenced LE motor recovery in patients. This application protocol with the possibility of progressive inclination towards vertical alignment prepared the patients for a smooth transition to the upright position, gradually increasing the load on the affected leg. As a result, a greater number of subjects who received progressive task-oriented training on the tilt table showed enhanced muscle activation in the LE and began to walk independently by the end of the intervention. Despite all the advantages of training with the application protocol in conjunction with the use of the supplementary tilt table, it is worth emphasizing that this rehabilitation program does not replace the daily applicable standard therapeutic treatment and serves only as an addition to it. From our view point, an important aspect of training is that progressive task-oriented training on the supplementary tilt table is accompanied by the active participation of the patient.

4.1. EMG analysis and clinical measures

This study used EMG analysis and clinical evaluations to investigate changes in functional performance and control of LE movements after progressive task-oriented training on the tilt table, respectively. By combining muscle activation measurements and clinical evaluations, it is possible to examine whether functional improvement is accompanied by a change in motor control (Desmurget and Grafton, 2000). Most studies report the composite scores of standardized tests such as the BI and FMS score rather than determining the changes in motor control or coordination of LE movements (Fugl-Meyer et al., 1975; Wade, 2000). In the clinical setting, impairment and disability is generally assessed by ordinal scales such as the BI and FMS score. The NIHSS is also one of the most commonly used scales for assessing stroke severity (McPeak, 1996). In the present study, we measured the LE muscle activity during stepping-like movements (3 consecutive steps) on the tilt table. Since stepping performance may be largely facilitated by providing treatment in the supine position on a tilt table, it was widely used as a special experimental design for rehabilitation procedures (Solopova et al., 2011a,b). We also chose an angle of 30° because it has been shown in previous studies that syncope was more likely to occur at an angle over 30° (Welch, 1970).

4.2. Crossed muscle activation gains in the affected LE

After a 3-week intervention period, we typically observed activity in the BF and MG muscles during knee flexion of the affected leg. Activity was also observed in TA and RF muscles during knee extension in the task-oriented training group on the tilt table compared to the other groups (Fig. 1). In particular, BF and MG muscles showed increased activity in the affected leg, partly due to increments in the activation amplitude of knee and hip muscles, while the other groups displayed no changes. Meanwhile, BF, MG, and RF muscle activities in the less-affected leg decreased significantly in the task-oriented training group on the tilt table. In the control group, either there were no changes in BF and MG muscle activities or the RF muscle activity increased. It seems that contralateral-side influences from the less-affected leg to the affected leg exist and the muscle activation is enhanced by task-oriented training. Previous studies have demonstrated the activation of the affected primary motor cortex during upper and lower limb movements on the less-affected side (Dragert and Zehr, 2013; Weiller et al., 1992). Dragert and Zehr (2013) reported that the unilateral dorsiflexor high-intensity resistance training on the less-affected side increases strength and motor output bilaterally following stroke. This finding is in agreement with our results. Another explanation for the observation is that short-term, task-oriented training on the tilt table with the less-affected side of the body causes "crossed" activation gains in the contralateral, untrained limb (Carroll et al., 2006; Munn et al., 2004). This finding also suggests a sort of non-paretic 'assistance' to the weak paretic limb using progressive task-oriented training. Therefore, we demonstrated that residual plasticity exists acute post-stroke and suggest clinical applications of the cross-education effect when training the affected side is not initially possible.

In our study, the task-oriented training group on the tilt table demonstrated a significantly greater increase in the activation of extensors and flexors of the affected leg muscle compared with the other groups. Since there were no significant differences in the baseline values among groups, the difference in the recovery rate was attributed to the progressive task-oriented training on

the tilt table. It should also be noted that there was a significant increase in the activation of extensor muscles of the less-affected leg, which was not significant in the other groups. The results obtained through our experiment showed that this approach is applicable beyond the tilt table. Tilt table exercises with functional activities such as one leg standing and task-oriented training may be more effective in improving the LE muscle activation, degree of performing activities of daily living, and LE function than the tilt table alone. The results of this study were in accordance with those in the study by Kuznetsov et al. (2013), which reported that robotic tilt table exercises with functional electrical stimulation is more effective in improving leg strength than tilt table alone. The robotic tilt table, Erigo (Hocoma AG, Volketswil, Switzerland), consists of a stretcher that can be tilted between 0° and 80° as well as foot plates with integrated springs for leg loading. However, loading of the legs on the robotic tilt table involves only passive or passive–active stepping movements (Kuznetsov et al., 2013) and the table is more expensive than a typical tilt table. Furthermore, training methods on the tilt table presented in our study can be easily applied to patients with stroke in clinical settings without the need for expensive robotic tilt tables and functional electrical stimulation. Thus, our results confirmed the effect of progressive task-oriented training on the tilt table in increasing the EMG levels in LE muscles after stroke.

4.3. The efficacy of the progressive task-oriented approach on the tilt table

According to international recommendations (Brainin et al., 2004), rehabilitation should start immediately after physical stability has been achieved, because the majority of the motor recovery occurs in the first few weeks after brain damage. In this study, intervention and tilt table training were started immediately after physical stability in patients was achieved. In episodes of stroke, the tilt table is generally one of the methods of early rehabilitation in the most acute and acute periods of stroke, and it is used to prevent muscle weakness and atrophy under the supervision of a physical therapist. However, more effective and standardized applications for the tilt table have not been presented. Therefore, we investigated whether task-oriented training on the tilt table after applying a knee belt, one of the tilt table accessories, can improve the LE muscle activation, degree of performing activities of daily living, and LE function in a different manner. As a result, progressive task-oriented training such as kicking a ball and target-matched exercises using the less-affected LE on the tilt table often provoked muscle activity in the affected leg. These results suggest that this type of training with knee and hip extension and flexion provides proprioceptive or supraspinal inputs to the spinal cord that increase motor recruitment compared to voluntary efforts (Dietz et al., 2002; Maegele et al., 2002) and may perhaps account for a better improvement in muscle activation and clinical scores in the task-oriented training group on the tilt table. The efficacy of the task-oriented approach for repeatedly practicing movement tasks associated with the patients' actual life has also been accepted in stroke rehabilitation (Yang et al., 2006). Its principles can be used to organize progressive task-oriented training using the tilt table protocol, with increasing clinical benefits in practicing the tasks. Therefore, based on our results, we suggest that the clinical benefits of progressive task-oriented training on a supplementary tilt table may be associated with familiarity of the tasks. Altogether, our results suggest that progressive task-oriented training on a supplementary tilt table may be a clinically feasible and promising approach for enhancing the LE muscle activation and function of patients with hemiparesis post-stroke, considering its clinical benefits and ease of application.

4.4. Study limitations

This study has several limitations. This study enrolled a small number of subjects, and did not assess biomechanical parameters such as joint movements and center of pressure. In addition, we were unable to analyze more parameters that characterize the changes in motor control as a result of the training in different conditions such as the changes in the muscle timing within the stepping cycles and co-contraction index between affected and less-affected LE. Therefore, definitive future studies with a large sample size and measurement of these biomechanical parameters from the LE muscles within the various stepping cycles would provide direct quantitative parameters for the functional phases of the gait cycle.

5. Conclusions

By demonstrating several significant findings in this study, we showed that progressive task-oriented training on the tilt table for early rehabilitation of stroke patients is an effective and standardized rehabilitation tool, which contributes to improvements in muscle activation, reduction of clinical symptoms, and enhanced performance in everyday life activities. It facilitates a smooth transition to walking in the vertical position, and thus, increases the effectiveness of rehabilitation and functional abilities of the patients.

Conflict of interest

The authors declare no conflict of interest.

Ethical approval

All procedures used in this study were approved by the Human Research Sciences of our local ethics committee.

Acknowledgements

Authors are grateful to all subjects involved in this study, as well as authors/publishers/editors of all articles, journals, and books reviewed and discussed for this study.

References

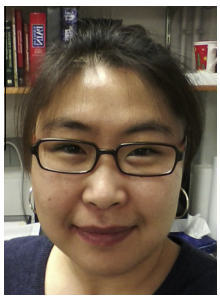
Aichner F, Adelwohrer C, Haring HP. Rehabilitation approaches to stroke. J Neural Transm Suppl 2002;63:59–73.
Brainin M, Olsen TS, Chamorro A, Diener HC, Ferro J, Hennerici MG, et al. EUSI executive committee; EUSI writing committee. Organization of stroke care: education, referral, emergency management and imaging, stroke units and rehabilitation, European stroke initiative. Cerebrovasc Dis 2004;17:1–14.
Brott T, Adams Jr HP, Olinger CP, Marler JR, Barsan WG, Biller J, et al. Measurements of acute cerebral infarction: a clinical examination scale. Stroke 1989;20(7):864–70.
Browse NL. Physiology and pathology of bedrest. Springfield. Charles C Thomas; 1965. p. 1–221.
Carroll TJ, Herbert RD, Munn J, Lee M, Gandevia SC. Contralateral effects of unilateral strength training: evidence and possible mechanisms. J Appl Physiol 2006;101(5):1514–22.
Criswell E. Cram's introduction to surface electromyography. Jones & Bartlett Publishers; 2010. p. 340–376.
Czell D, Schreier R, Rupp R, Eberhard S, Colombo G, Dietz V. Influence of passive leg movements on blood circulation on the tilt table in healthy adults. J Neuroeng Rehabil 2004;25(1):4.
Daniels L, Worthingham C. Muscle testing: techniques of manual examination. 6th ed. Philadelphia: W.B. Saunders; 2003. p. 120–52.
Desmurget M, Grafton S. Forward modeling allows feedback control for fast reaching movements. Trends Cogn Sci 2000;4:423–31.
Dietz V, Müller R, Colombo G. Locomotor activity in spinal man: significance of afferent input from joint and load receptors. Brain 2002;125(Pt 12):2626–34.
Dragert K, Zehr EP. High-intensity unilateral dorsiflexor resistance training results in bilateral neuromuscular plasticity after stroke. Exp Brain Res 2013;225(1):93–104.

Eliasziw M, Young SL, Woodbury MG, Fryday-Field K. Statistical methodology for the concurrent assessment of interrater and intrarater reliability: using goniometric measurements as an example. Phys Ther 1994;74(8):777–88.
Faul F, Erdfelder E, Lang AG, Buchner A. G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. Behav Res Methods 2007;39(2):175–91.
Fugl-Meyer AR, Jääskö L, Leyman I, Olsson S, Steglind S. The post-stroke hemiplegic patient. A method for evaluation of physical performance. Scand J Rehabil Med 1975;7(1):13–31.
Furlan R, Porta A, Costa F, Tank J, Baker L, Schiavi R. Oscillatory patterns in sympathetic neural discharge and cardiovascular variables during orthostatic stimulus. Circulation 2000;101(8):886–92.
Hainsworth R, Al-Shamma YM. Cardiovascular responses to upright tilting in healthy subjects. Clin Sci (Lond) 1988;74(1):17–22.
Hermens HJ, Freriks B, Merletti R, Hagg GG, Stegeman D, Blok J, et al. SENIAM 8: European recommendations for surface ElectroMyoGraphy. Enschede, NL: Roessingh Research and Development; 1999.
Kautz SA, Brown DA. Relationships between timing of muscle excitation and impaired motor performance during cyclical lower extremity movement in post-stroke hemiplegia. Brain 1998;121(Pt 3):515–26.
Kisner C, Colby LA. Therapeutic exercise: foundations and techniques. 5th ed. Philadelphia: F.A. Davis Company; 2007. p. 147–223.
Kuznetsov AN, Rybalko NV, Daminov VD, Luft AR. Early poststroke rehabilitation using a robotic tilt-table stepper and functional electrical stimulation. Stroke Res Treat 2013;2013:946056.
Luk'ianov AL, Shamalov NA, Ivanova GE, Skvortsova VI. Passive tilting in patients in the acute period of cerebral stroke. Zh Nevrol Psikhiatr Im S Korsakova 2010;110(4 Suppl 2):29–35.
Maegele M, Müller S, Wernig A, Edgerton VR, Harkema SJ. Recruitment of spinal motor pools during voluntary movements versus stepping after human spinal cord injury. J Neurotrauma 2002;19(10):1217–29.
McPeak LA. Physiatrichistory and examination. In: Braddom R, editor. Physical medicine and rehabilitation. W.B. Saunders Company; 1996. p. 3–42.
Munn J, Herbert RD, Gandevia SC. Contralateral effects of unilateral resistance training: a meta-analysis. J Appl Physiol 2004;96(5):1861–6.
Nudo RJ. Postinfarct cortical plasticity and behavioral recovery. Stroke 2007;38:840–5.
Rensink M, Schuurmans M, Lindeman E, Hafsteinsdóttir T. Task-oriented training in rehabilitation after stroke: systematic review. J Adv Nurs 2009;65(4):737–54.
Riberholt CG, Thorlund JB, Mehlsen J, Nordenbo AM. Patients with severe acquired brain injury show increased arousal in tilt-table training. Dan Med J 2013;60(12):A4739.
Shumway Cook A, Woollacott MH. Motor control: translating research into clinical practice. 3rd ed. Philadelphia: Lippincott Williams & Williams; 2007. p. 147–470.
Solopova IA, Selionov VA, Zhvanskiĭ DS, Grishin AA. Mutual influences of upper and lower extrimities during cyclic movements. Fiziol Cheloveka 2011a;37(4):55–64.
Solopova IA, Tihonova DY, Grishin AA, Ivanenko YP. Assisted leg displacements and progressive loading by a tilt table combined with FES promote gait recovery in acute stroke. NeuroRehabilitation 2011b;29(1):67–77.
Sylos-Labini F, Ivanenko YP, Maclellan MJ, Cappellini G, Poppele RE, Lacquaniti F. Locomotor-like leg movements evoked by rhythmic arm movements in humans. PLoS One 2014;9(3):e90775.
Tennant LM, Maly MR, Callaghan JP, Acker SM. Analysis of muscle activation patterns during transitions into and out of high knee flexion postures. J Electromyogr Kinesiol 2014;24(5):711–7.
Wade DT. Measurement in neurological rehabilitation. Oxford University Press; 2000.
Wagner HN. Orthostatic hypotension. Bull Johns Hopkins Hosp 1959;105:322–59.
Weiller C, Chollet F, Friston KJ, Wise RJ, Frackowiak RS. Functional reorganization of the brain in recovery from striatocapsular infarction in man. Ann Neurol 1992;31(5):463–72.
Welch R. Tilt-table therapy in rehabilitation of the trauma patient with brain damage and spinal injury. Nurs Clin North Am 1970;5(4):621–30.
Yang YR, Wang RY, Lin KH, Chu MY, Chan RC. Task-oriented progressive resistance strength training improves muscle strength and functional performance in individuals with stroke. Clin Rehabil 2006;20(10):860–70.

Chang-Yong Kim M.Sc., PT received his master of science in physical therapy at the Daejeon University in 2012. From 2012 to 2014, he worked as a physical therapist at Samsung Medical Center. He is currently studying the course of Ph.D. in the Korea University and is a member of the Korean Physical Therapy Association. His research interests include musculoskeletal response in physical therapy interventions, lumbopelvic regions, and ankle instability.

Jung-Sun Lee B.H.Sc., OT is an occupational therapist and is a member of the Korean Occupational Therapy Association. She received a B.H.Sc. degree in Occupational Therapy from Konyang University in 2012 and is currently studying the course of M.Sc. in the Korea University. From 2012 to 2014, she worked as an occupational therapist at Samsung Medical Center. She is primarily interested in Developmental Psychology, Occupational therapy for children, Occupational therapy evaluation, Psychiatry, and Sensory Integration.

Hyeong-Dong Kim Ph.D., PT was awarded a degree in Business Administration from the Korea University, a Bachelor of Science from the Taegu University, Master of Health Science in Physical Therapy from the University of Florida and his Ph.D. from the University of Florida. He is a professor in the department of physical therapy at Korea University, Seoul, South Korea. His research interest includes an understanding of movement dysfunction with a biomechanical approach.



Junesun Kim Ph.D., PT is an associate professor in the department of physical therapy at Korea University, Seoul, South Korea. She received a B.Sc. degree in Physical Therapy from Shamyook University and a M.Sc. degree in Life Science & Biotechnology from Korea University. Once, she worked as a physical therapist at Sinchon Severeance Hospital, Yonsei University. She completed her Ph.D. degree in the Department of Neurobiology at Korea University. She is primarily interested in the pain control, spinal cord injury, Physical therapy evaluation, and neuroscience.

In-Hak Lee Ph.D., PT is a professor in the department of physical therapy of Daejeon Health and Sciences College, Daejeon, South Korea. He received his B.Sc. degree in Physical Therapy and his M.P.H. degree from Chungnam national University, and a Ph.D. in the Republic of Korea from Daegu Haany University. He is currently a member of the Korean Physical Therapy Association. His research interest focuses on the evaluation of normal and pathologic motor functions of the shoulder and spine using comprehensive diagnostic approaches, primarily of the electromyography techniques.